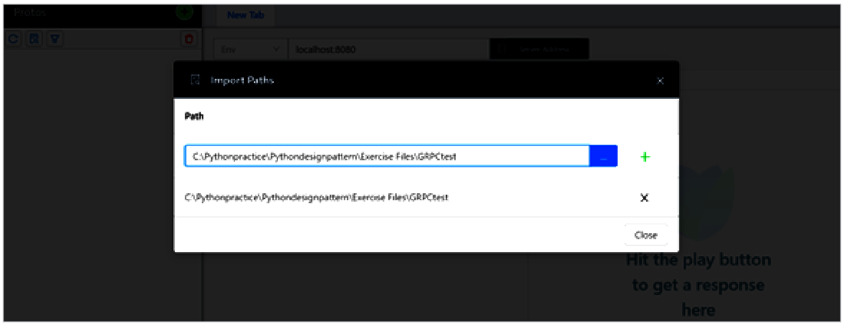

Figure 2: Adding the *path* of the folder

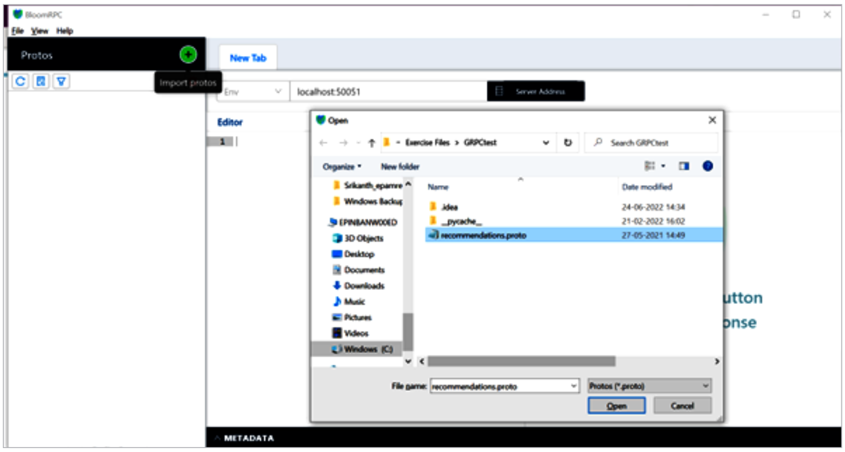

Figure 3: Importing the *protofile*

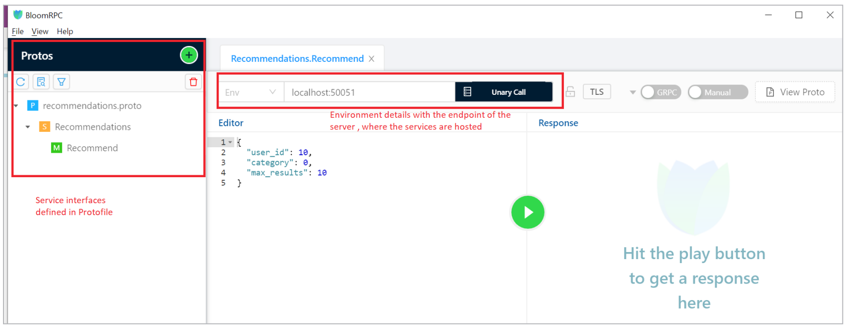

Figure 4: Service descriptions post *protofile* import

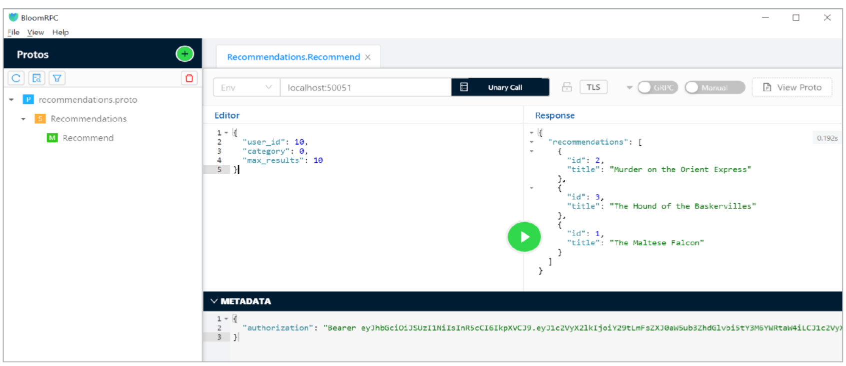

Figure 5: Details of the response received

You need to provide the topmost folder containing the *protobuf* files.

Importing the *protobuf* file: To see the service interfaces we have to test, we need to import the *protobuf* files into the tool, as shown in Figure 3.

After the *protofile* is imported, you will find a service description, as shown in Figure 4. You will find the service interfaces are described in the left side pane of the tool and the message details in the right. Also, we will see the endpoint details, which need to be provided by the user depending on the environment to be used. Services might be hosted on different ports, so we need to gather the details for our test. We can save the environment details with a name for future use.

Once you hit the *play* button, you will get a response to the message (refer Figure 5). We can further validate the response messages as per our requirements.

Sending metadata/auth information along with the message: Many services may require certain authentication or other metadata information to be sent along with the request. We send such information in the metadata field, as shown in Figure 5.

This article has introduced BloomRPC as one of the tools to validate gRPC calls. However, there are other tools like Postman and grpcui that provide comprehensive support for validating gRPC services. The validation of gRPC is similar to that of Rest services, except for the difference in bi-directional and streaming data. While testing, we should understand the message types and how they are structured to achieve effective validation. Happy learning! END

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